



EcoCampus

AI-powered Smart & Green Campus Platform

Building a Closed-Loop Circular Economy for Sustainable Campuses



Team 1 – GCNB

Green · Code · No Bug

University of Computer Studies, Mandalay (UCSM)

THE PROBLEM

When graduation arrives or semesters end, why do tons of fully functional items end up in campus trash cans instead of find new owner.



Resource Waste

Books, clothing, and electronics are often discarded into landfills.



Disposal Confusion

Students lack clear guidance on reusing, reusing, recycling, or recycling, or donating items.



Lack of Motivation

Eco-friendly actions lack visible incentives and feedback.

VALUABLE RESOURCES → WASTED POTENTIAL

OUR UNIQUE INSIGHT

Sorting waste is only half the battle—the true key is an incentive-driven circular economy.

Beyond Trash Sorting



From disposal to lifecycle care

Value-Driven Habit



Green actions = real value

Hyper-Local Ecosystem



Trusted student-only network.

We don't just sort waste — we unlock value.



OUR INNOVATION

Not Just Another Second-Hand Market:
The Green Points Advantage.



THE SMART ENGINE: AI INTEGRATION

AI Integration Beyond a Basic Chatbot

Visual Object Recognition

Instantly identifies uploaded items via computer vision.



Instantly identifies uploaded items via computer vision.

Smart Eco-Routing

AI analyzes item condition and recommends the optimal disposal path, Reuse, Recycle, or Donate.



Dynamic Valuation



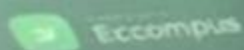
Suggests fair Green Point values based on item condition and category.

Intelligent Demand Matching

Connects idle resources with students who actively need them.



Connects idle resources with students who actively need them.



EcoCampus Ecosystem

One Campus. One Green Economy.

EcoCampus connects environmental actions with a gamified campus marketplace. Together, we create value, community, and impact.



1



Campus Registration

Students sign up with Student ID, Email, Password, and Year / Batch.

New users receive Initial Exchange Coins.

2



Green Points

Earn Green Points through everyday environmental actions.

- Recycling
- Waste Sorting
- Green Activities
- Volunteering
- AI Recycling Scan

Green Points represent your environmental contribution.

3



Year / Batch Leaderboard

Green Points are accumulated by Year / Batch and displayed on a ranked leaderboard.

Competition Period
→ Ranking →
Friendly Competition

4



Exchange Coins

Convert Green Points into Exchange Coins.



Exchange Coins are used in the Marketplace.

Converting Green Points into Exchange Coins does not affect your Leaderboard score.

5



Reuse Marketplace

Buy, sell, or trade used items with Exchange Coins.

- Textbooks
- Stationery
- Clothing
- Daily-use Items
- ...and more

THE ECOCAMPUS LOOP



ACT
Take green actions.

EARN
Earn Green Points.

EXCHANGE
Convert to Exchange Coins.

REUSE
Exchange or reuse items.

IMPACT
Create a greener campus.

ACT → EARN → EXCHANGE → REUSE →
IMPACT

Smart & Connected Campus

Where AI Meets the Green Campus

EcoCampus integrates AI-powered tools, community connections, and real-time data to build a smarter, more sustainable campus.



AI Recycling Classification

Point, shoot, and upload an image of a recyclable item. Gemini AI automatically classifies the item and awards Green Points instantly.



Ask AI

Ask anything about the environment, campus life, or projects. Gemini AI is here to help you.

Environmental Questions

Campus Questions

Project / General Questions

Ask → Understand → Take Better Action

Campus & Yearly Chat Feeds

Connect and communicate within your campus community.

Year / Batch Feed

Chat exclusively with students from the same Year / Batch.

Campus Feed

A general feed for all campus users to share and engage.

Share ideas · Discuss · Organize · Stay Connected

Admin Dashboard

Admin can monitor and manage the platform with a comprehensive dashboard.

 Registered Users

 Listed Items

 Recycling Images

Real-time data · Smart insights · Better decisions



 Powered by Gemini (gemini-3.6-flash)

 Building a greener campus through technology, community, and continuous impact.



TRADITIONAL MARKETPLACE

Trade with Money. Value is limited.



TRADITIONAL SECOND-HAND MARKETPLACE



Transaction Medium
Real Money



Value Source
Money



Purpose
Buy / Sell to save or make money



User Behavior
Buy → Sell



Incentive Mechanism
No environmental incentive



Environmental Impact
Limited impact



AI Assistance
Usually not a core function



Resource Flow
Second-hand circulation only



Core Value
Economic Value

TRADITIONAL FLOW



Have an item



Sell it



Receive money



Buy another item



ECO CAMPUS GREEN POINT MARKETPLACE

Trade with Green Actions. Value is unlimited.



ECOCAMPUS GREEN POINT MARKETPLACE



Transaction Medium
Green Points



Value Source
Environmental actions (Recycle, Clean up, Green activities, etc.)



Purpose
Promote sustainability & resource reuse



User Behavior
Act → Earn → Exchange → Reuse



Incentive Mechanism
Green Points reward for positive actions



Environmental Impact
High positive impact on campus & environment



AI Assistance
AI Scan, AI Advice, Smart Recommendations



Resource Flow
Green action → Reuse → Reduce waste → Circular economy



Core Value
Environmental + Social Value



ECOCAMPUS FLOW



Do something
green



Earn
Green Points



Exchange in
Marketplace



Item gets a
second life



Create bigger
environmental impact



BUILD A GREENER
CAMPUS TOGETHER



REDUCE WASTE
IN CAMPUS



ENCOURAGE POSITIVE
ENVIRONMENTAL ACTIONS



STRENGTHEN CAMPUS
COMMUNITY



CREATE A SUSTAINABLE
FUTURE

WHY WE STAND OUT: SMART & GREEN

Dual-Core Value Proposition

Where is Smart?

- Automated visual verification of eco-actions.
- AI-driven resource routing and price recommendation.
- Real-time supply and demand optimization.



Real-time supply and demand optimization.



Where is Green?

- Maximizes campus waste sorting and material recovery.
- Extends product lifecycles and reduces unnecessary consumption.
- Quantifies campus-wide carbon footprint reduction.



Quantifies campus-wide carbon footprint reduction.

**DON'T JUST THROW IT AWAY.
GIVE IT ANOTHER LIFE.**

ECOCAMPUS

Smart Campus · Circular Value · Greener Future

THANK YOU.